Related work, v2: set completeness in LLMs and search agents

Daryl Alfaro • November 5, 2026 • focus: novelty and publication potential of the expanded benchmark

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Bottom line. Showing that search agents miss items is *no longer novel*. Since mid-2025, several benchmarks ask agents for complete sets and find large gaps:

- **WideSearch:** best success rate of about 5%.
- **DeepSearchQA:** the top agent is fully correct on 66% of prompts despite an F1 of 82.
- **WANDR:** performance drops as the number of target records grows.

None of them, however, is *diagnostic*. None controls answer-set size or candidate-set size. None asks the same questions with full context, with distributed documents, and with search. None measures whether a missed item was never found or was found and then dropped. Those are the gaps the expanded benchmark should own. I checked each entry below against its abstract and, for the closest ones, the paper body (2026-10-27).

1. Search agents asked for complete sets (closest to the expanded scope)

- **WideSearch** (Wong et al., ByteDance Seed; ICLR 2026). [link] 200 real-user tasks (English and Chinese) in which agents collect many verifiable facts from the live web into a Markdown table, often 100–1,000 cells per task.
  - *Metrics:* success requires a perfect table; row- and item-level F1 give partial credit.
  - *Results:* most systems are near 0% success and the best reaches 5%. Recall is consistently below precision. With 128 attempts, item-level F1 nears 80 while table-level success stays under 20.
  - *Not reported:* performance by the number of rows required.
- **DeepSearchQA** (Gupta et al., Google DeepMind; 2026). [link] 900 hand-written prompts across 17 fields, many with set answers, grounded in the open web.
  - *Metrics:* each answer is categorised as fully correct, partially correct, "correct with extraneous answers" or fully incorrect.
  - *Results:* the top agent scores F1 81.9 but is fully correct on only 66.1% of prompts, which the authors call the "last mile problem"; this matches our recall vs. exact-set gap.
  - *Failure modes named:* premature stopping (missing the long tail) and hedging (padding the answer to inflate recall). More test-time compute raises full correctness from 67% to 86%.
  - *Not reported:* performance by answer-set size.
- **WANDR** (Polshkov et al., Perplexity; Aug 2026). [link] 500 tasks requiring many entities (breadth) and evidence for each (depth), from dozens to thousands of records.
  - *Scoring:* judges re-fetch the cited pages; there is no static gold set.
  - *Results:* best soft F1 0.36 and hard F1 0.13. Performance "drops as target volume and hierarchy depth increase". Bottlenecks: incomplete discovery, missing enrichment and incomplete evidence.
  - *Closest to our answer-size finding,* but not controlled and not separated from task difficulty.
- **DeepWideSearch** (Lan et al.; 2025). [link] 220 questions combining multi-hop depth with many items. Agents average 2.39% success. Failure modes: lack of reflection, over-reliance on internal knowledge, insufficient retrieval, context overflow.
- **LiveDRBench** (Java et al.; 2025). [link] 100 deep-research tasks, including entity enumeration over scientific and public-interest topics. F1 ranges from 0.02 to 0.72; the best system scores 0.55.
- **AutoResearchBench** (Xiong et al.; Apr 2026). [link] Its "Wide Research" task asks for the complete set of papers meeting given conditions, with an unknown number of answers. The best model reaches 9.3% IoU. **This overlaps our planned scientific-publications domain.**

2. Fixed corpora that separate retrieval from reasoning

- **Total Recall QA (TRQA)** (Rafiee et al.; 2026). [link] Fixed corpora (5.9M Wikipedia passages; 3.2M documents about synthetic e-commerce products) with answers built from a knowledge base. Each query needs *every* relevant entity, but asks for one aggregated value (a count, mean, sum and so on), not the set.
  - *Set sizes:* 9–22 relevant entities on average, at most 50.
  - *Results:* even with oracle retrieval GPT-5.2 reaches only 56.8 exact match, and over 90% of oracle errors are reasoning errors. Agents' number of sub-queries does not grow with the number of gold entities.
  - *Not reported:* performance by set size.
  - *Relevance:* closest to setting C's "searchable collection with hidden ground truth", but its output is a single number, not a set.
- **BrowseComp-Plus** (Chen et al.; ACL 2026). [link] Replaces the live web with a fixed 100,195-document corpus with human-verified evidence and hard negatives, so the retriever and the model can be evaluated separately (GPT-5: 55.9% with BM25, 70.1% with a strong embedding retriever). Answers are short, single facts.

3. Full-context and structured completeness (our current baseline)

- **HoloBench** (Maekawa et al.; ICLR 2025). [link] Database rows written out as long text; five query types; varies context length and information density; LLM-judge scoring with partial credit. Does not control answer-set size.
- **Oolong** (Bertsch et al.; 2025). [link] Long-context aggregation with scalar answers.
- **LongProc** (2025). [link] HTML→TSV row extraction scored by row-level F1, with no selection conditions.
- **LOFT** (Lee et al.; 2024). [link] Finds SQL-like compositional reasoning hard for long-context models.

4. Many-answer questions, entity retrieval and recall from memory

- **QAMPARI** (2023), **RoMQA** (2023) and **QUEST** (2023): list answers over real corpora, where retrieval, knowledge and reading failures are confounded.
- **NERetrieve** (2023): retrieving *all* mentions of an entity type; standard retrievers fall well short.
- **Empty Shelves or Lost Keys?** (Calderon et al.; ICML 2026): failures of facts stored in a model's memory are mostly *recall* failures rather than missing knowledge. Relevant to our memory-only setting.

5. The two pilot domains

- **Scientific publications: crowded.**
  - *AutoResearchBench* (above) asks for complete paper sets.
  - **PaSa** (ByteDance Seed and Peking University; ACL 2025). [link]: a crawler-plus-selector agent for "comprehensive" paper search, with real-query evaluation.
  - **LitSearch** (EMNLP 2024). [link]: 597 queries, but mostly retrieval of specific papers rather than complete sets.

  To be distinct, our tasks should use conditions with *verifiable complete answers* from a fixed snapshot (e.g. OpenAlex): explicit (venue and year), relational (co-authorship, citations) and semantic (topic from abstracts).
- **Magic: The Gathering: open.** Existing LLM work on Magic covers gameplay and drafting (e.g. UrzaGPT) and rules reasoning. TCG-Bench uses an invented card game. I found no completeness benchmark, so this is a clean domain, with a complete, versioned card database to use as ground truth.

Where the expanded benchmark fits

	Access setting	Output	Size effect measured	Access vs. reasoning separated	Set failure decomposition
WideSearch	live web	table of entities	no	no	failure categories
DeepSearchQA	open web	set or single	no	no	premature stop / hedging
WANDR	live web	records + evidence	qualitative	no	bottleneck categories
DeepWideSearch	live web	table	no	no	failure categories
AutoResearchBench	search	paper set	no	no	–
TRQA	fixed corpus	one number	no	yes (oracle retrieval)	accumulated recall
BrowseComp-Plus	fixed corpus	short answer	–	yes	–
HoloBench	full context	mixed	density, length	–	no (LLM judge)
This work	**full context, distributed, search; same answer sets**	**sets**	**controlled: answer size × candidate size, item-level**	**yes (identical sets across settings)**	**never found / found but omitted / wrong / stopped early**

Novelty and publication potential

What is still open, and what our preliminary results already support:

1. **Controlled scaling.** No benchmark reports completeness as a function of answer-set size with candidate-set size held fixed, or tests the independent-error baseline. Our analysis (`analysis/item_level.md`) shows that each item's chance of being missed rises with answer size (odds ratio 1.50 per doubling, 95% interval 1.24–1.83) and with table size (3.0, 2.25–4.03), and that misses clump within answers. The answer-size results reported elsewhere (WANDR) are not controlled.
2. **Attributing failures to access, reasoning or output.** Asking the same questions, with the same answer sets, with full context, with distributed documents and with search would show where items are lost. BrowseComp-Plus and TRQA separate retrieval from reasoning only for short or aggregated answers.
3. **Set-level failure decomposition, measured.** DeepSearchQA and WideSearch *name* premature stopping and incomplete retrieval. None measures how much of the evidence the agent saw ended up in its answer on a corpus with hidden ground truth.
4. **Graded condition types** (explicit, relational, semantic) with authoritative, versioned ground truth.

Main risk. Reviewers may say WideSearch and DeepSearchQA already cover this. The paper must frame itself as a *diagnostic, controlled* benchmark that explains *why and where* items are lost. It must not frame itself as one more hard search benchmark, and it should cite and compare against WideSearch, DeepSearchQA, WANDR and HoloBench directly.

Reusing WideSearch. WideSearch has released its 200 tasks and gold tables under CC0 (public domain), but not per-task model results: its leaderboard reports only aggregate scores. So its published results can't be re-analysed by answer size. Its tasks and gold tables, however, could serve as real-world entity sets of known size for our own controlled runs.

Venue fit: NeurIPS Datasets & Benchmarks or ICLR, given the diagnostic benchmark; SIGIR or CIKM resource tracks if the retrieval and search analysis leads.

Not yet verified: details beyond the abstracts for DeepWideSearch, LiveDRBench and AutoResearchBench (metric definitions, corpus); the ICLR-version WideSearch numbers; whether Scryfall's terms allow redistributing a data snapshot.